

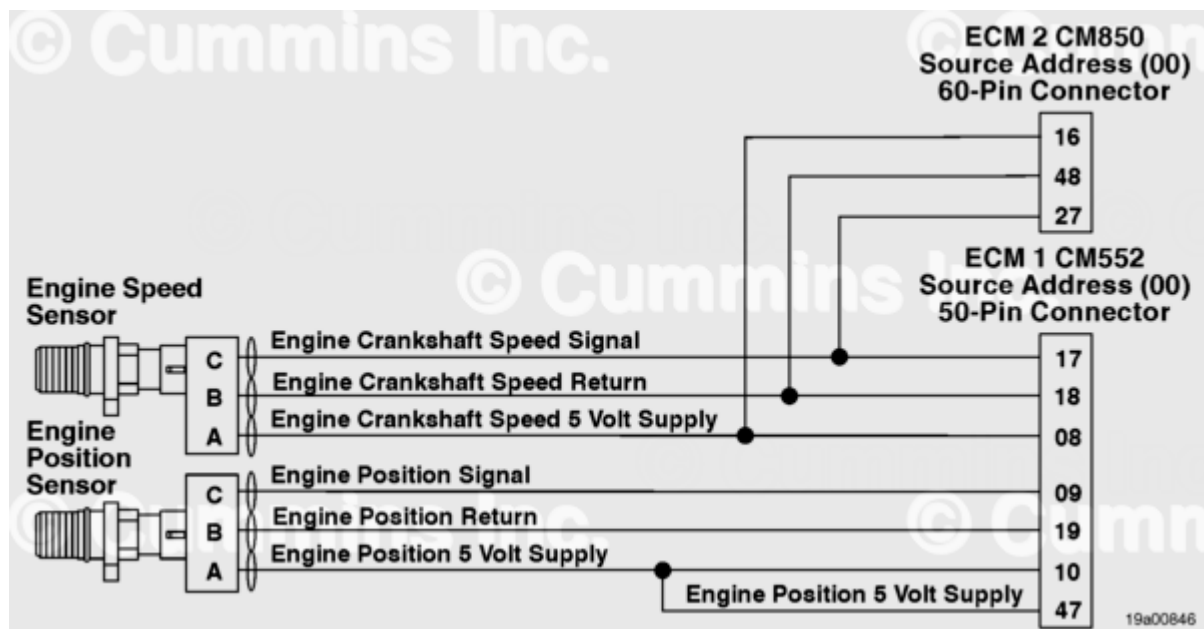
Fault Code: 115

Engine Magnetic Speed/Position Lost Both of Two Signals - Data Erratic, Intermittent, or Incorrect

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Overview

Codes	Reason	Effect
Fault Code: 115 PID(P): P190 SPN: 612 FMI: 2/2 Lamp: Red SRT:	Both signals failed to be detected at the engine position sensor circuit.	The engine will shut down or will not start.



Engine Crankshaft Speed Sensor Circuit

Circuit Description

The crankshaft position and camshaft position sensors are Hall effect type sensors. The engine control module (ECM) provides a 5 volt supply to the position sensor and a return circuit. As the teeth on the crankshaft speed ring or the dimples in the back of the camshaft gear move past the

position sensor, a signal is generated on the position sensor signal circuit. The ECM interprets this signal and converts it to an engine speed. A missing tooth on the crankshaft gear is used by the ECM to determine the position of the engine.

Component Location

The engine speed sensor and the engine position sensor are located in the flywheel housing.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the crankshaft position and camshaft position signal inputs to the ECM are missing or reversed.

Action Taken When The Fault Code Is Active

- The generator set controller displays a shutdown fault immediately when the diagnostics runs and fails.
- Engine torque will be reduced if the engine is operated for an extended period of time with this fault active.

Conditions For Clearing The Fault Code

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the shutdown indicator immediately after the user presses the reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Crankshaft position sensor and the camshaft position sensor wiring harness connectors are reversed.
- Damaged or loose connectors.
- Malfunctioning or damaged engine wiring harness.

